

Review

Fusion energy: a sustainable pathway to meeting future global energy demands

Gaurav Katoch¹ · Gazal Sharma² · Mohammed Alaghbari³ · Amit Ved⁴ · Jatinder Kaur⁵ · Asha Rajiv⁶ · Abhayveer Singh⁷ · Yashpal Yadav⁸ · Mohammad Iqbal Khairandish⁹

Received: 4 November 2024 / Accepted: 10 February 2025

Published online: 31 March 2025

© The Author(s) 2025 [OPEN](#)

Abstract

Since the 1930s, scientists have worked to replicate nuclear fusion as a reliable energy source. Fusion energy holds immense promise as a sustainable and virtually limitless solution for power generation. Fusing light nuclei, particularly deuterium and tritium, yields fusion processes that release considerably more energy than fossil fuels and nuclear fission sources. This study examines the merits of fusion energy, emphasising its sustainability advantages, such as little environmental effect and plentiful fuel resources. Recent developments in confinement technologies, like tokamaks and stellarators, have made considerable progress in achieving sustained fusion events. **Nonetheless, obstacles persist, especially in maintaining plasma stability and attaining net-positive energy production.** This analysis emphasises the necessity of ongoing research and worldwide cooperation to surmount these obstacles and realise the potential of fusion energy for global energy requirements.

Keywords Fusion energy · Tokamak · Plasma physics · Magnetic confinement · Deuterium–tritium · Sustainability

1 Introduction

The global energy sector is undergoing a profound transformation driven by rising energy demands, industrial growth, and the urgent need for sustainable solutions. This shift emphasizes adopting renewable sources, advanced technologies, and innovative methods to address climate change and energy security challenges. **Energy consumption is likely to be elevated due to anticipated population growth and rapid industrialisation, necessitating the formulation of strategies to address this need with minimal ecological repercussions [1].** The International Energy Agency (IEA) forecasts a 30% increase in worldwide energy use by 2040 [2]. This results from economic expansion

✉ Gaurav Katoch, gaurav.18681@rayatbahrauniversity.edu.in; ✉ Mohammad Iqbal Khairandish, iqbal.khairandish@gmail.com; Gazal Sharma, gazal.sharma@rayatbahrauniversity.edu.in; Mohammed Alaghbari, mohammed.alaghbari@asu.edu.bh; Amit Ved, amit.ved@marwadieducation.edu.in; Jatinder Kaur, jatinder.cgctc@gmail.com; Asha Rajiv, asha.rajiv@jainuniversity.ac.in; Abhayveer Singh, abhayveer_singh@outlook.com; Yashpal Yadav, yashpal.yadav@nimsuniversity.org | ¹Department of Physics, Rayat Bahra University, Punjab, Mohali, India. ²Department of Research, Rayat Bahra University, Mohali, Punjab, India. ³College of Administrative Sciences, Applied Science University, Eker, Kingdom of Bahrain. ⁴Department of Electrical Engineering, Faculty of Engineering and Technology, Marwadi University Research Center, Marwadi University, Rajkot, Gujarat 360003, India. ⁵Department of Electronics and Communication Engineering, Division of Research and Innovation, Chandigarh Engineering College, Chandigarh Group of Colleges Jhanjeri, Mohali, Punjab 140307, India. ⁶Department of Physics and Electronics, School of Sciences, JAIN (Deemed to be University), Bangalore, Karnataka, India. ⁷Centre for Research Impact and Outcome, Chitkara University Institute of Engineering and Technology, Chitkara University, Rajpura, Punjab 140401, India. ⁸Department of Electrical, Electronics and Electric Vehicle Engineering, NIMS Institute of Engineering and Technology, NIMS University Rajasthan, Jaipur, India. ⁹Civil Engineering, Faculty of Jami University, Herat, Afghanistan.



and enhanced living standards in developing countries. This need intensifies as climate change deteriorates, mainly because of fossil fuel consumption [3]. The principal energy sources are the burning of coal, oil, and natural gas, which constitute 70% of world CO₂ emissions [4]. To meet global energy and climate targets, especially the ones specified in the Paris Agreement it is important to embrace a shift toward low-carbon and renewable energy sources [5].

Solar, wind, and water energy sources are expanding rapidly as key components of the renewable energy landscape. However, they face significant challenges that hinder their scalability [6]. These include intermittent availability and geographical limitations, making them difficult to harness consistently in certain regions. While wind and solar energy are vital for a sustainable future, their inherent unpredictability means that reliable energy storage systems or well-thought-out backup strategies are essential to maintain a stable and continuous power supply [7]. This hinders their ability to serve as the primary energy source in many locations, particularly during periods of high demand or unfavourable weather conditions [3]. Renewable energy infrastructure is generally more cost-effective than traditional fossil fuel systems. However, it is presently lacking the capability to fully substitute the steady and reliable base-load energy supplied by fossil fuels [1]. This limitation has driven the pursuit of innovative and sustainable energy solutions. Fusion energy emerges as a particularly promising option, with the potential to revolutionise global energy production by providing an abundant, clean, and dependable power source [8]. It employs light atomic isotopes, such as deuterium and tritium, to generate energy similar to the sun [9]. In contrast to nuclear fission, which results in radioactive waste, atomic fusion yields helium, an inert and harmless gas [10]. Additionally, seawater may contain deuterium, the primary fuel for fusion, suggesting its potential for continuous use [11]. The process of locating fossil energy resources is quite challenging.

Fusion technology can conserve more energy than both fossil fuels and renewable sources. According to various studies, one gram of deuterium–tritium fuel might produce as much energy as ten tons of coal without producing dangerous gases like CO₂ [1]. Nuclear fission poses meltdown risks and difficulties in long-term waste management, while fusion offers a safer and more sustainable approach to producing significant energy [8]. Light atomic nuclei combine to produce fusion energy, which can replace nuclear fission and fossil fuels. Fusion, unlike fission, produces less radioactive waste and uses saltwater (deuterium) as a sustainable fuel source [11]. Chemical energy sources exhibit significantly lower energy density than fusion [10]. Conventional chemical reactions, like the combustion of fossil fuels (e.g., coal, oil, natural gas) or biofuels, play a significant role in energy production. For example:

- The combustion of coal produces approximately 24 MJ/kg.
- Oil yields about 44 MJ/kg, and natural gas provides 55 MJ/kg under standard conditions.

These chemical energy sources are reliant on the breaking of molecular bonds during combustion, which inherently limits their energy density. In comparison, fusion derives energy from nuclear reactions, specifically from the combination of atomic nuclei, which releases energy on the order of millions of MJ/kg, as shown in Table 1. To put this into perspective:

- When deuterium and tritium (in a 50:50 ratio) fuse, they create an energy density of about 330,000,000 MJ/kg, which is much higher than that of chemical sources.
- It would take tens of millions of kilograms of coal or oil to burn to get the same amount of energy as 1 kg of deuterium–tritium fuel.

The stark contrast in energy density highlights fusion energy's potential as a clean and sustainable energy source for the future. The comparison is clearly demonstrated in Table 1, which also illustrates that the focus on Uranium-235 as the primary fission energy source is due to its widespread application and the maturity of its technology in nuclear power generation. Alternative isotopes such as plutonium-239 have distinct energy densities and challenges, which make them less common in commercial applications.

While the potential of fusion energy is vast, recent advancements in alternative and complementary approaches deserve significant attention. For instance, the National Ignition Facility has demonstrated key milestones in inertial confinement fusion (ICF), including meeting the Lawson criterion for ignition [14]. Additionally, hybrid concepts like magneto-inertial fusion (MIF), which integrate magnetic and inertial confinement principles, are paving new pathways toward achieving energy production goals [15, 16].

The exploration of advanced fusion cycles and alternative systems, such as low-radioactive and hybrid fusion approaches, is continuing to gain traction in addressing key limitations [17, 18]. These approaches emphasise enhanced safety, reduced environmental impacts, and scalability, solidifying fusion's role in clean energy initiatives. This comparison

Table 1 Comparison of energy sources in terms of energy density, environmental impact, and scalability

Energy source	Energy density (MJ/kg)	CO ₂ emissions (kg CO ₂ /kWh)	Current global usage (%)	Projected future growth (%)	Cost per kWh (₹)	Countries leading in development/usage	Environmental impact	Timeline
Fusion (deuterium–tritium)	330,000,000	0	0	10	4.16	US, China, EU, Japan (ITER)	Minimal, long-term waste concerns (tritium)	Ongoing research, commercial by 2050
Fission (Uranium-235)	88,000,000	0.005	10	7	9.14	US, Russia, France, China, India	Low, radioactive waste management needed	1940's (Manhattan Project), power since 1950's
Coal	24	0.91	27	15	4.99	China, India, US, Germany, Australia	High pollution, land degradation	1760s (Industrial Revolution)
Oil	44	0.77	33	20	8.31	Saudi Arabia, US, Russia, Canada, Iraq	High pollution, oil spills, greenhouse gases	1859 (First commercial oil well)
Natural gas	55	0.45	24	28	5.82	US, Russia, Qatar, Iran, Canada	Moderate, methane emissions	Early twentieth century
Solar	0.6	0.03	3	40	3.32	China, US, Germany, India, Japan	Minimal, land use concerns	1950s (Commercial solar power development)
Wind	0.5	0.02	5	35	4.16	China, US, Germany, Denmark, India	Minimal, impact on birds and land use	1980s (modern wind power development)

Adapted from [12, 13], and projections based on fusion adoption data

underscores fusion energy's capacity to satisfy global energy demands while alleviating various environmental issues, and establishing it as a viable alternative to chemical energy sources in the long term. The analysis also emphasises the imperative of progressing fusion technology to attain a sustainable and scalable energy future.

Fusion energy has great potential, but there are significant technological difficulties to overcome. For instance, creating and maintaining the necessary conditions for fusion requires a large amount of energy [11]. To help make fusion power plants a reality by 2050, multinational projects like the International Thermonuclear Experimental Reactor (ITER) are being developed. These projects aim to demonstrate how fusion energy can be harnessed for practical use [9]. In Fig. 1 it has been clearly demonstrated, that fusion energy might substantially aid in achieving the CO₂ emission reduction targets set by the Paris Agreement for 2050. It might be employed in conjunction with renewable energy sources [5].

While fusion energy holds significant potential as a clean and virtually limitless power source, its practical utilization remains a major challenge. According to numerous studies, the principal obstacles are excessively high pressure and extremely elevated temperatures (exceeding 100 million degrees Celsius) [10]. A substantial amount of energy is required to maintain these conditions, and the current technology is still in its early stages of development [11]. The International Thermonuclear Experimental Reactor and other large-scale fusion experiments evaluate the viability of nuclear fusion. Fusion research is progressing toward its goal of delivering clean, unlimited, and sustainable energy by the mid-twenty-first century. This is enabled by transnational partnerships, governmental funding, and innovative ideas from the corporate sector [11].

This review paper examines the advancements in fusion energy research and the challenges facing its potential as a global energy source. In this work, we have analyzed fusion physics, recent developments, and global activities to evaluate whether fusion can effectively address international energy and environmental challenges. We have also examined the impact of fusion on global CO₂ emissions and its alignment with the goals of the Paris Agreement [5]. This analysis highlights how fusion could contribute to the generation of a future free from carbon-based energy sources.

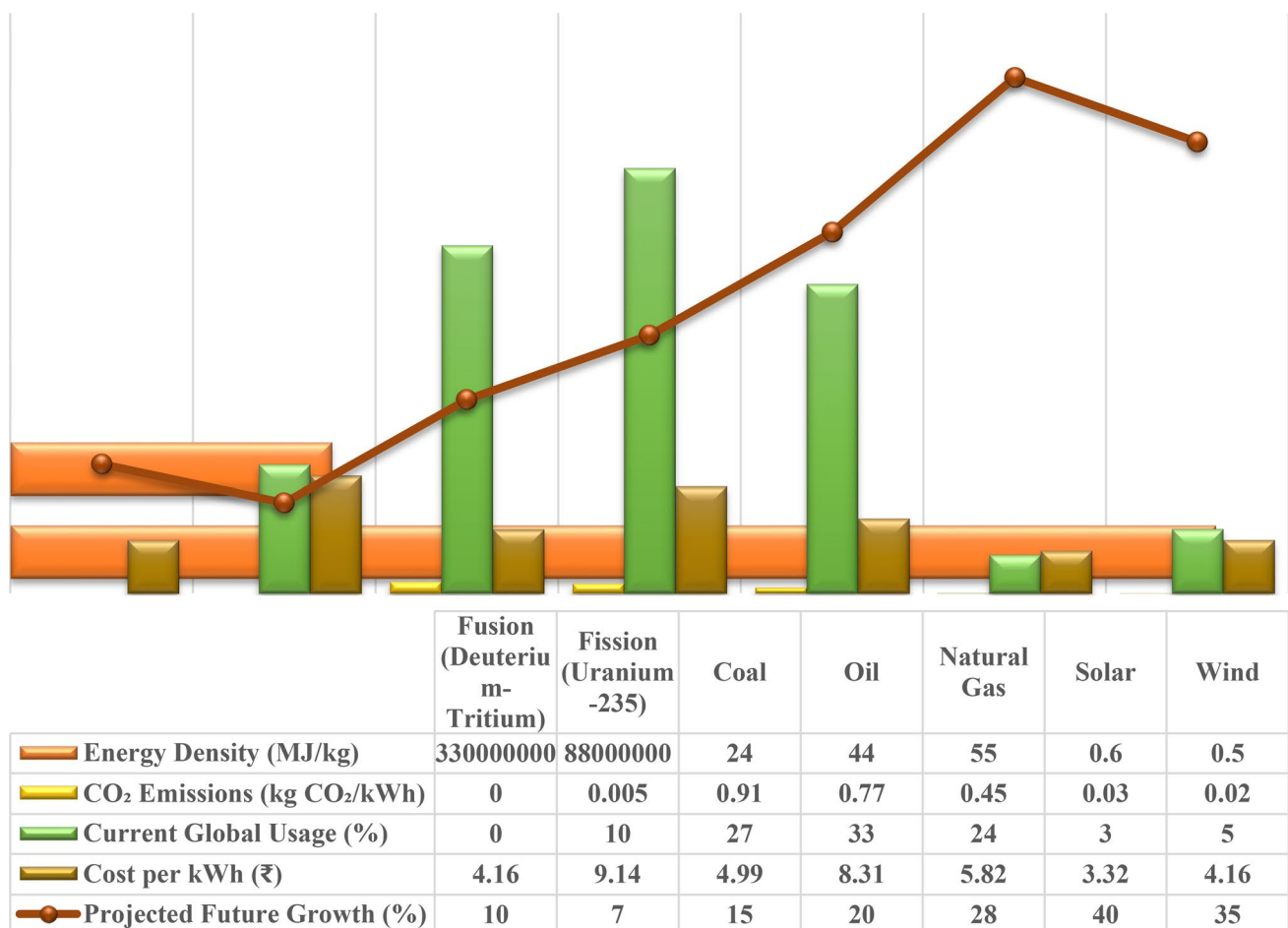


Fig. 1 Comparing energy density and CO₂ emissions of major energy sources

2 Energy landscape and the case for fusion energy

The global energy landscape remains heavily reliant on fossil fuels, which account for over 80% of total primary energy consumption. This reliance has significant environmental consequences, with fossil fuel combustion contributing approximately 70% of global CO₂ emissions. The International Energy Agency (IEA) projects a 30% increase in global energy demand by 2040, driven by economic growth and population expansion in developing regions. Such trends underscore the urgent need to transition to cleaner and more sustainable energy sources to mitigate climate change and environmental degradation.

2.1 The role of renewables and their limitations

Renewable energy sources like solar, wind, and hydropower have experienced remarkable growth over the past decade. Their advantages, including low operational costs and reduced greenhouse gas emissions, make them critical components of the energy transition. However, renewables face significant challenges:

- **Scalability:** Expanding renewable infrastructure often requires substantial investment and land use, which may not be feasible in all regions.
- **Intermittency:** Solar and wind energy depend on weather conditions, leading to variability in power generation and necessitating energy storage solutions.
- **Geographical constraints:** Hydropower and wind farms require specific geographic conditions, limiting their deployment in certain areas.

These limitations hinder the ability of renewables to consistently meet base-load energy demands, particularly in industrialised economies with high and continuous energy requirements.

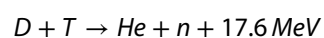
2.2 The case for fusion energy

Fusion energy offers a promising alternative to the limitations of current energy sources, showcasing several significant advantages. Firstly, the high energy density of fusion is remarkable, with fusion reactions producing energy orders of magnitude greater than chemical reactions. Just one gram of fusion fuel is capable of generating as much energy as eight metric tons of oil. Furthermore, minimal waste is generated by fusion, with the long-lived radioactive byproducts associated with fission being avoided. The primary byproduct, helium, is inert and environmentally harmless. Additionally, fusion is considered sustainable, as abundant resources like deuterium from water and tritium, which can be bred from lithium, are utilized, ensuring a virtually limitless supply of fuel for the future.

International initiatives like ITER and SPARC are paving the way for the commercialisation of fusion energy. ITER, the world's largest experimental fusion reactor, aims to demonstrate the feasibility of sustained fusion reactions on a large scale. A compact high-field fusion project, SPARC, focuses on achieving net energy gain in a smaller and potentially more economically viable reactor. These efforts underscore the global commitment to harnessing fusion as a clean, safe, and sustainable energy source for future generations.

3 Physics of fusion energy

The primary fusion reaction of interest involves deuterium (D) and tritium (T), resulting in helium and a neutron while releasing approximately 17.6 MeV of energy:



This reaction requires extremely high temperatures, around 100 million degrees Celsius and pressures to overcome the repulsive forces between atomic nuclei [19]. The basic mechanism of fusion energy production hinges on several necessities, including high temperature, high pressure, and effective confinement [20]. High temperatures provide the kinetic energy necessary for the nuclei to overcome their electrostatic repulsion. At these temperatures, matter exists in a plasma state, where electrons are separated from nuclei, creating conditions conducive to fusion. Additionally, sufficient pressure is needed to increase the likelihood of collisions between the nuclei. In stars, this

pressure is naturally provided by gravitational forces, but on Earth, it must be artificially created through advanced confinement techniques [21].

To maintain the conditions required for fusion energy, the plasma must be effectively confined to prevent it from coming into contact with the reactor walls, which would cause it to cool and lose the necessary heat for fusion. This is achieved through magnetic confinement in devices like tokamaks and stellarators, which utilise strong magnetic fields to keep the plasma stable and prevent it from coming into contact with the reactor walls. The strength and configuration of these magnetic fields are critical factors influencing confinement times, with stronger magnetic fields typically leading to longer confinement and more stable plasma, thereby increasing the likelihood of successful fusion reactions [22]. A diagram of a tokamak, with labelled components, is illustrated in Fig. 2, highlighting the intricate design required for effective magnetic confinement.

Several limiting factors pose significant challenges to the production of fusion energy. Plasma instability is considered one of the most significant issues, as stable plasma at extreme temperatures is difficult to maintain. The plasma loses its confinement due to disruptions, resulting in a loss of the temperature and pressure required for fusion. Additionally, the materials used in fusion reactors must withstand extreme temperatures and neutron bombardment, creating challenges for the reactors' longevity and safety. As current materials degrade over time, continuous research is needed to discover more durable and resilient alternatives for reactor construction.

A fundamental challenge is ensuring that fusion produces more energy than it consumes as the energy produced by fusion must exceed the energy required to initiate and sustain the reaction [23]. This requires advancements in confinement techniques and reactor design to optimise energy gain. Fuel availability also presents a significant concern, while deuterium is abundant in seawater, tritium is comparatively scarce and must be bred from lithium within the reactor. A sustainable supply of tritium must be ensured for the long-term viability of fusion energy.

The physics of fusion energy predominantly involves a complex interplay of elevated temperatures, pressures, and sophisticated confinement technologies. This is essential to render fusion energy a viable energy source. Figure 3 presents a schematic picture of the fundamental process behind the generation of fusion energy, specifically depicting a D–T fusion reaction [19].

4 Recent advances in fusion technology

The International Thermonuclear Experimental Reactor, currently built in France, represents the largest global collaboration in fusion energy research [12, 24]. This initiative involves over 35 countries, including the European Union countries, the United States, the Russian Federation, and China, aiming to demonstrate the feasibility of sustained nuclear fusion as a scalable, carbon-free energy source. ITER targets a power gain factor (Q) of 10 by producing 500 MW of

Fig. 2 Diagram of a Tokamak with labelled components (Li, S., Jiang, H., Ren, Z., and Xu, C. 2014. "Optimal Tracking for a Divergent-Type Parabolic PDE System in Current Profile Control." *Abstract and Applied Analysis* 2014: 940965. <https://doi.org/10.1155/2014/940965>.)

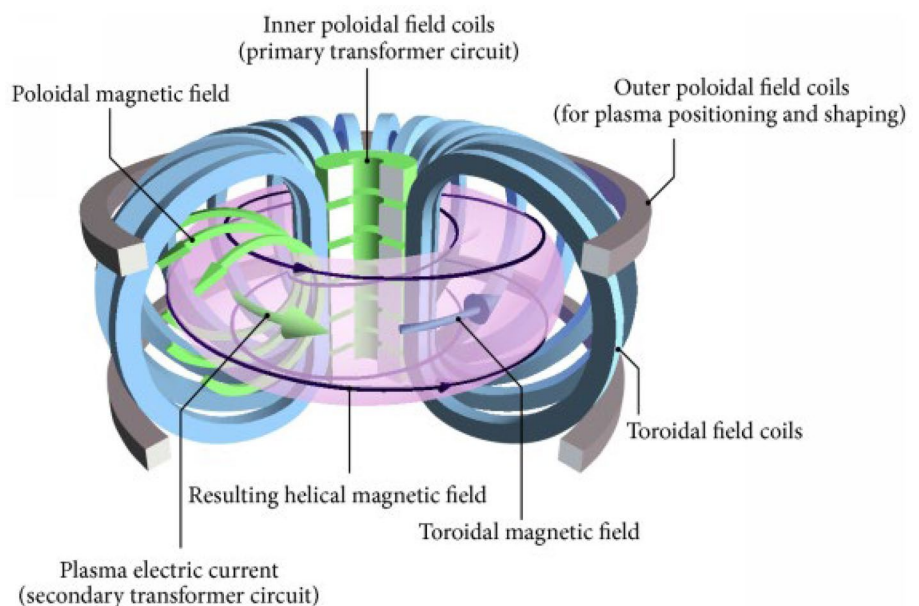
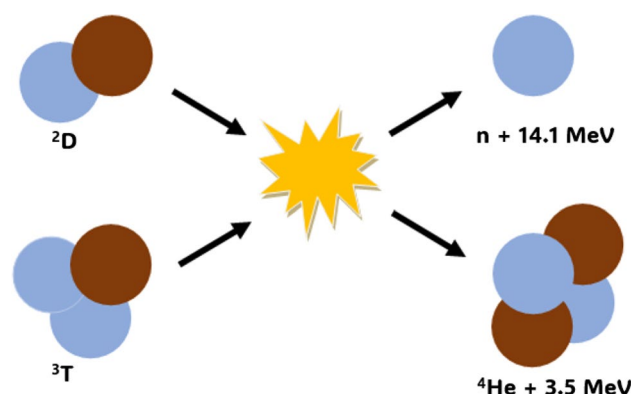


Fig. 3 Schematic of a D–T fusion reaction



fusion power with 50 MW of input, which is a significant step toward practical fusion energy [12, 22]. In addition to ITER, private-sector projects like the Commonwealth Fusion Systems' SPARC focus on using high-temperature superconductors to create stronger magnetic fields in smaller tokamak designs [14, 25]. SPARC aims to reach breakeven or beyond by the early 2030s, providing a complementary path to ITER's larger-scale experiments [26].

4.1 Magnetic field control

One of the breakthroughs in fusion research has been the development of high-temperature superconducting (HTS) magnets, which allow for much stronger magnetic fields than traditional superconductors [26]. These magnets enable more compact tokamak designs, potentially reducing the overall size and cost of fusion reactors while maintaining or even improving their performance. Stronger magnetic fields improve the stability of plasma and extend confinement times, allowing the plasma to maintain the necessary temperatures for fusion reactions to occur [21].

Research into controlling plasma turbulence has also made significant progress. By using advanced computational models and machine learning algorithms, researchers can predict and control turbulence within the plasma [27], thereby reducing energy losses. These models help to optimise plasma performance, manage disruptions, and mitigate instabilities that could otherwise lead to energy losses or damage to reactor components [28].

4.2 Materials science

Fusion reactors face extreme conditions, including intense neutron fluxes that can slowly degrade materials over time [19, 26]. Fusion reactors operate under extreme conditions, including intense neutron fluxes that can gradually degrade materials over time. In response, significant advancements in materials science have led to the development of radiation-resistant alloys and composites capable of enduring these harsh environments. Due to its high melting point and exceptional resistance to neutron bombardment, tungsten is regarded as an ideal material for the divertor, the component responsible for managing the exhaust heat generated by the plasma. Additionally, beryllium has been selected as a first-wall material due to its low atomic number, which minimises plasma contamination along with favourable thermal conductivity conditions. These materials play a crucial role in optimising the performance and longevity of fusion reactors [29, 30].

Additionally, researchers are exploring self-healing materials that can recover from neutron-induced defects over time and increase the longevity of reactor components. The precise development of complex geometries that optimise cooling and heat dissipation in fusion reactor designs is made possible by advancements in modern manufacturing, such as 3D printing (additive manufacturing) [31]. These efforts are particularly relevant to the advancements in magnetic confinement devices, as detailed in Table 2, which compares the main types of magnetic confinement technologies.

Table 2 Comparison of MAIN magnetic confinement technologies

Technology/device	Stability	Efficiency	Magnetic field strength	Current examples	Recent innovations
Tokamaks	High	Moderate	5–10 Tesla	ITER, SPARC	HTS magnets, machine learning-based turbulence control
Stellarators	Moderate	High	3–5 Tesla	Wendelstein 7-X	Optimisation of magnetic coil design
Spheromaks	Low	High	Variable	Experimental	Compact design, cost-effective reactors
Advanced Tokamaks	High	High	10–15 Tesla	Experimental (SPARC)	High-confinement modes (H-mode), self-organised plasmas

5 Sustainability implications of fusion energy

Fusion Energy offers a transformative solution for reducing greenhouse gas emissions. Fusion addresses one of the major challenges of nuclear energy by not producing long-lived radioactive waste, which is a concern with nuclear fission [32]. **The primary byproduct of D–T fusion is helium**, an inert gas that does not contribute to atmospheric pollution [33, 34]. Moreover, fusion's reliance on non-fossil fuels makes it an essential technology for global decarbonisation efforts [12].

Fusion reactors have the potential to run continuously, offering stable base-load power that supports the variable nature of renewable sources such as solar and wind [28]. This feature positions fusion as a reliable backbone of a low-carbon energy grid, particularly in countries with high energy demand [12].

Fuel efficiency: The fuel for fusion, deuterium, is abundant in seawater, making it a sustainable option for large-scale energy production [20, 35]. **Tritium, though rarer, can be bred in reactors using lithium blankets, creating a self-sustaining fuel cycle** [25, 36]. This minimises the need for external fuel sources and enhances the long-term viability of fusion as a global energy solution.

In terms of energy yield, 1 g of deuterium–tritium fuel can release as much energy as burning about 10 tons of coal, significantly reducing the environmental impact associated with energy extraction and use [31]. Figure 4 and Table 3 illustrate the possible influence of fusion energy on global CO₂ emissions, commencing in 2020 and extending to 2050. The data covers three main categories:

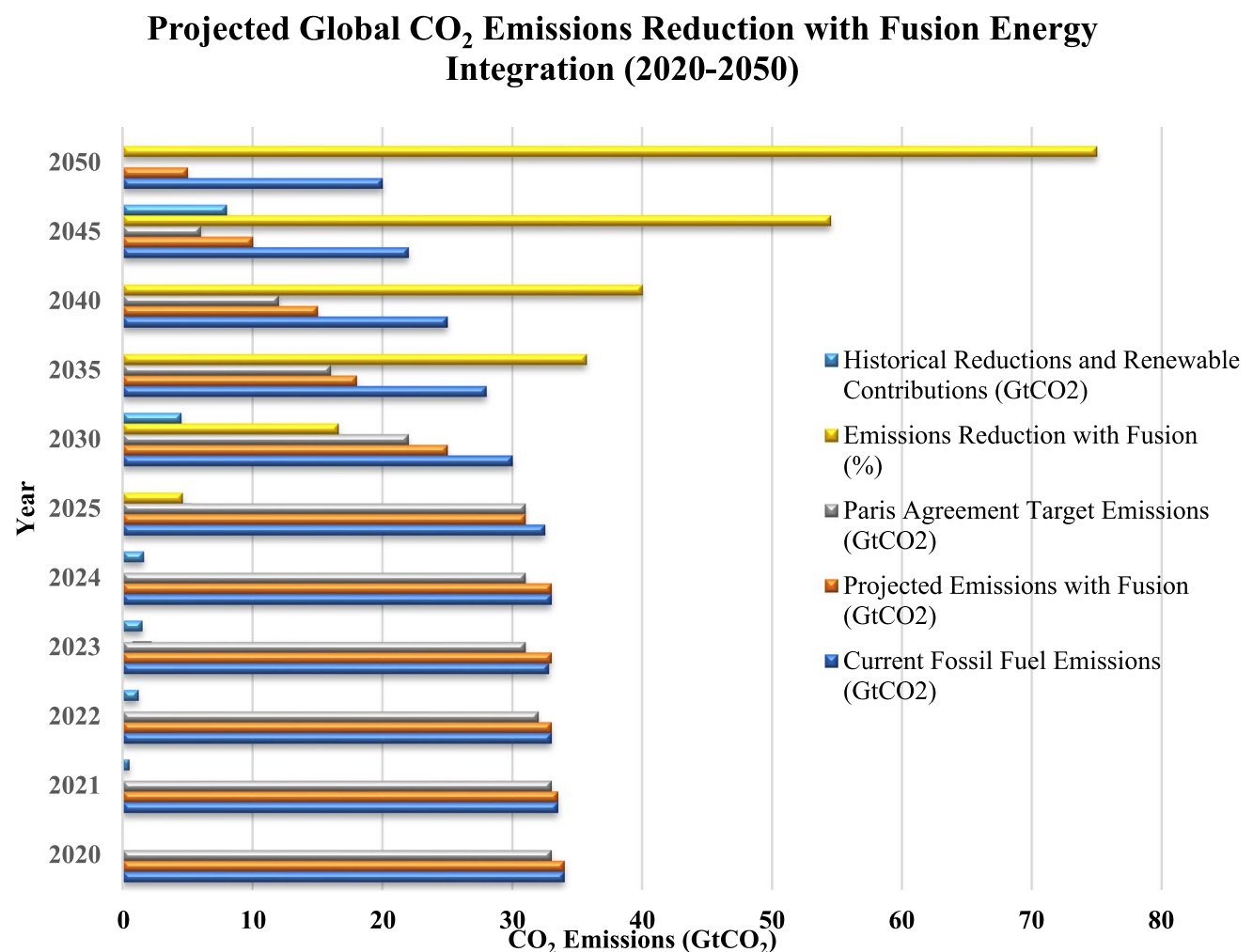


Fig. 4 Comparison of current fossil fuel emissions and projected reductions with fusion energy technology, contrasted against Paris Agreement targets. The chart illustrates the gradual adoption of fusion energy, beginning in 2030, and its significant role in reducing global CO₂ emissions by 75% by 2050, alongside contributions from renewable energy and carbon capture technologies

Table 3 Projected global CO₂ emissions from 2020 to 2050 with fusion energy integration and renewables contribution

Year	Current fossil fuel emissions (GtCO ₂)	Projected emissions with fusion (GtCO ₂)	Paris agreement target emissions (GtCO ₂)	Emissions reduction with fusion (%)	Historical reductions and renewable contributions (GtCO ₂)
2020	34	34	33	0	1 (due to renewables + pandemic impact)
2021	33.5	33.5	33	0	0.5
2022	33	33	32	0	1.2
2023	32.8	33	31	0	1.5
2024	33	33	31	0	1.6
2025	32.5	31	31	4.6	2 (with increasing renewables)
2030	30	25	22	16.6	4.5
2035	28	18	16	35.7	6.2 (fusion adoption starts)
2040	25	15	12	40	7 (fusion + renewables dominating)
2045	22	10	6	54.5	8
2050	20	5	0	75	10 (fusion dominant, another tech)

1. Current fossil fuel emissions: Historical and projected levels of emissions based on ongoing fossil fuel usage without fusion energy.
2. Projected emissions with fusion: This column represents the expected emissions reductions if fusion energy is introduced, assuming gradual implementation starting around 2030.
3. Paris agreement target emissions: These values illustrate the emissions trajectory needed to meet the Paris Agreement goals of keeping global temperature rise well below 2 °C and aiming for 1.5 °C.

The figure shows that without new technology, global CO₂ emissions from fossil fuels will steadily decrease due to renewable energy growth but will not meet the Paris targets until fusion and additional technologies (like carbon capture and storage) are implemented. By 2035, fusion energy is projected to start significantly reducing emissions, with a more pronounced impact by 2040. By 2050, fusion technology could cut emissions by 75%, with only 5 GtCO₂ remaining globally, compared to 33 GtCO₂ in 2020. This optimistic scenario assumes rapid advancements in fusion energy and other supporting technologies like renewables and Carbon Capture and Storage (CCS). The Paris Agreement targets indicated on the chart provide a critical benchmark for global climate action (Table 4).

6 Challenges and future directions

A significant challenge in fusion energy research is achieving the “breakeven” point, where the energy generated from fusion reactions equals or surpasses the energy input required to sustain them [19–22]. The Lawson Criterion defines the necessary conditions for this achievement, emphasising the need for sufficient temperature, particle density, and confinement time [37]. While ITER aims for a power gain factor (Q) of 10, SPARC targets a gain of around 20, potentially reaching the “ignition” stage, where the reaction becomes self-sustaining [21, 30, 37].

Table 4 Status of global fusion projects

Project	Location	Current status	Target plasma energy	Breakeven date estimate	Key milestones
JET	UK	Operational	16 MW	2025	World's first D–T fusion experiment
ITER	France	Under Construction	500 MW	2035	The first plasma is expected in 2025, with full power by 2035
SPARC	USA	Design Phase	100 MW	2030	First to use HTS magnets for compact design
KSTAR	South Korea	Research	30 MW	2028	Record plasma confinement time of 20 s

Technological barriers:

1. Plasma instabilities: Addressing plasma instabilities like edge localised modes (ELMs) is essential for ensuring continuous operations. These instabilities can cause sudden energy losses or damage to reactor components. Techniques like resonant magnetic perturbations (RMPs) and real-time feedback control are being developed to mitigate these effects.
2. Heat management: The high temperatures inside fusion reactors pose challenges for managing heat loads, especially in the divertor region, which must withstand extreme heat while allowing the exhaust of helium ash. Novel cooling techniques, such as liquid metal divertors and advanced heat-resistant materials are being explored to improve reactor longevity.
3. **Tritium breeding and handling: Developing efficient tritium breeding** and safe handling systems is vital for the self-sufficiency of fusion reactors. Lithium blankets are designed to absorb neutrons and generate tritium, but optimising this process for industrial-scale reactors remains a challenge.

Global fusion research receives support from initiatives like the EU's Horizon 2020, the U.S. Department of Energy's ARPA-E, and private investments from companies like CFS and TAE Technologies [33]. These collaborations aim to expedite fusion research and reduce reactor development costs. Experts project that commercial fusion energy could be viable by 2050 if the current technical and financial challenges are addressed [21, 22, 25].

7 Conclusion

Fusion energy presents a promising pathway to achieving a sustainable and nearly inexhaustible energy supply. The deuterium–tritium (D–T) fusion reactions are considered a viable alternative to fossil fuels, significantly reducing their environmental impact while meeting global energy demands. In contrast to traditional nuclear energy, fusion generates much less long-lived radioactive waste and does not emit carbon dioxide, positioning it as a transformative clean energy source with an energy density far exceeding that of fossil fuels.

Recent advancements in plasma confinement technologies, such as tokamaks and high-temperature superconducting magnets, along with machine learning approaches for turbulence management, have accelerated progress in fusion research. International collaborations, particularly through major programs like ITER and private sector initiatives like SPARC, underscore the growing momentum towards achieving net-positive energy generation. However, it is important to note that considerable technological challenges remain. These include achieving the extreme conditions necessary for fusion (temperatures reaching 150 million degrees Celsius) and developing the advanced materials and technologies required to sustain these conditions over extended periods of time [22, 27].

From an economic standpoint, fusion's high energy density suggests the potential for cost-effective power generation in the long term. However, the substantial initial investments required for infrastructure and reactor development remain a significant challenge [20, 33]. Nonetheless, with increasing public funding, private investment, and growing interest in renewable energy markets, fusion could emerge as a competitive energy source, particularly as a reliable base-load power complementing intermittent renewables like wind and solar.

An important practical consideration is the potential for fusion energy to work synergistically with other renewable sources, such as hydrogen, solar, and wind. For example, hydrogen production through fusion-powered electrolysis could facilitate the creation of a low-carbon hydrogen economy with large-scale potential. Moreover, the integration of fusion with renewable energy grids could address intermittency issues while benefiting from fusion's high energy density and reliability. While fusion may have higher initial costs compared to other energy sources, long-term benefits, including reduced dependence on fossil fuels and minimal greenhouse gas emissions, suggest that fusion has the potential to play an important role in the transition to clean energy. Nonetheless, more detailed cost–benefit analyses are needed to assess the true financial implications of such integrated systems.

The potential of fusion energy to reduce carbon emissions and support global climate objectives is significant [34]. By providing a virtually limitless and clean energy source, fusion could reshape the global energy landscape, especially for rapidly industrializing economies. Despite the challenges, ongoing collaboration between public and private sectors and international initiatives indicate that fusion may play a crucial role in global energy strategies by the mid-twenty-first century.

In conclusion, while fusion energy holds immense promise, significant technological, economic, and regulatory hurdles remain to be addressed before it can reach its full potential. However, when combined with other renewable energy sources, fusion could play a pivotal role in dramatically reducing carbon dioxide emissions by 2050. This could significantly advance international climate initiatives, such as the Paris Agreement, and offer a sustainable, long-term solution to the world's growing energy needs. As research and development continue, fusion energy may ultimately help pave the way toward a cleaner, more resilient global energy system [24, 31].

Author contributions Gaurav Katoch led data collection, formal analysis, and drafting of the initial manuscript. Gazal Sharma provided supervision, and resources, and contributed to the review and editing of the manuscript. Mohammed Alaghbari supported data collection and contributed to writing the manuscript. Amit Ved was involved in formal analysis and visualisation. Jatinder Kaur assisted with resources and administration. Asha Rajiv contributed to data validation and formal analysis. Abhayveer Singh provided support in visualisation and manuscript review. Yashpal Yadav assisted with data interpretation and review. Mohammad Iqbal Khairandish contributed to the review and editing of the manuscript.

Funding Not applicable.

Data availability The data described in this manuscript can be freely and openly accessed.

Declarations

Ethics approval and consent to participate Not applicable.

Competing interests The authors declare no competing interests.

Open Access This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by-nc-nd/4.0/>.

References

1. Rahman MM, et al. The future of energy: transitioning to low-carbon solutions. *Renew Energy*. 2022;180:123–35. <https://doi.org/10.1016/j.renene.2021.09.040>.
2. International Energy Agency. 2020. World Energy Outlook 2020. IEA Publications. <https://www.iea.org/reports/world-energy-outlook-2020>.
3. Zhou Y, et al. The impact of energy transition on CO₂ emissions. *Environ Sci Policy*. 2021;121:71–9. <https://doi.org/10.1016/j.envsci.2021.03.012>.
4. Intergovernmental Panel on Climate Change. Climate change 2021: the physical science basis. In: Contribution of working group I to the sixth assessment report of the intergovernmental panel on climate change, vol. 1. Cambridge University Press; 2021.
5. UNFCCC. Paris Agreement. United Nations Framework convention on climate change. 2015. <https://unfccc.int/>.
6. Sachs J, et al. Sustainable Development Report 2020: the sustainable development goals and COVID-19 includes the SDG index and dashboards. Cambridge: Cambridge University Press; 2021. <https://doi.org/10.1017/9781108992411>.
7. Li H, et al. Energy density of fusion: a comprehensive review. *Energy*. 2022;243:122–32. <https://doi.org/10.1016/j.energy.2021.122132>.
8. Kramer GJ, et al. Fusion energy: a sustainable path forward. *Nat Energy*. 2022;7(4):321–9. <https://doi.org/10.1038/s41560-022-01010-9>.
9. ITER Organisation. 2021. ITER: the World's largest fusion experiment. ITER organisation reports. <https://www.iter.org/>. Accessed 10 Oct 2023.
10. Smith J. Nuclear fusion: the next frontier. *J Energy Environ Science*. 2020;13(9):2458–68. <https://doi.org/10.1039/D0EE01245C>.
11. Garcia AI, et al. Sustainable fusion energy: the future of power generation. *Energy Rep*. 2021;7:856–67. <https://doi.org/10.1016/j.egyr.2021.01.003>.
12. ITER Organisation. 2023. ITER progress report 2023. ITER organisation reports. <https://www.iter.org/magazine>.
13. Guidi G, Violante AC, Iuliis SD. Environmental impact of electricity generation technologies: a comparison between conventional, nuclear, and renewable technologies. *Energies*. 2023;16(23):7847. <https://doi.org/10.3390/en16237847>.
14. Abu-Shawareb H, Acree R, Adams P, Adams J, Addis B, Aden R, Adrian P, Afeyan BB, Aggleton M, et al. Lawson's criterion for ignition exceeded in an inertial fusion experiment. *Phys Rev Lett*. 2022;129(7): 075001. <https://doi.org/10.1103/PhysRevLett.129.075001>.
15. Ryzhkov SV. Magneto-inertial fusion and powerful plasma installations (a review). *Appl Sci*. 2023;13(11):6658. <https://doi.org/10.3390/app13116658>.

16. Thio YCF, Hsu SC, Witherspoon FD, Cruz E, Case A, Langendorf S, Cook A. Plasma-jet-driven magneto-inertial fusion. *Fusion Sci Technol*. 2019;75(7):581–98. <https://doi.org/10.1080/15361055.2019.1598736>.
17. Ryzhkov SV. Low radioactive and hybrid fusion: A path to clean energy. *Sustain Cit Soc*. 2015;14:313–5. <https://doi.org/10.1016/j.scs.2014.05.003>.
18. Bradshaw AM, Hamacher T, Fischer U. Is nuclear fusion a sustainable energy form? *Fusion Eng Des*. 2011;86:2770–3. <https://doi.org/10.1016/j.fusengdes.2010.11.040>.
19. Bohm D, Gross EP. Theory of plasma oscillations. a. origin of medium-like behavior. *Phys Rev*. 1949;75(12):1851–64. <https://doi.org/10.1103/PhysRev.75.1851>.
20. Smirnov V. Fusion physics: fundamentals and applications. *Fusion Sci Technol*. 2009;56(4):1–8. <https://doi.org/10.13182/FST09-A1004>.
21. Raman A, Anoma M, Zhu L, et al. Passive radiative cooling below ambient air temperature under direct sunlight. *Nature*. 2014;515:540–4. <https://doi.org/10.1038/nature13883>.
22. Aymar R, et al. Overview of the ITER project. *Fusion Sci Technol*. 2002;41(1):1–14. <https://doi.org/10.13182/FST02-A120>.
23. Mueller D. Magnetic confinement in plasma physics. *J Plasma Phys*. 2013;79(1):1–24. <https://doi.org/10.1017/S0022377812000821>.
24. Arnoux JJ, Sutter G, List G, Molinari A. Friction experiments for dynamical coefficient measurement. *Adv Tribol*. 2011;2011:613581. <https://doi.org/10.1155/2011/613581>.
25. Takahashi H. Fusion reactor materials. *Nucl Fusion*. 2010;50(8): 084009. <https://doi.org/10.1088/0029-5515/50/8/084009>.
26. Bauer SE, et al. Historical (1850–2014) aerosol evolution and role on climate forcing using the GISS ModelE2.1 contribution to CMIP6. *J Adv Model Earth Syst*. 2020;12(8):e2019MS001978. <https://doi.org/10.1029/2019MS001978>.
27. Bruzzone S, et al. The ITER project and its challenges. *Fusion Eng Des*. 2017;123:59–64. <https://doi.org/10.1016/j.fusengdes.2017.01.006>.
28. Menard JE, et al. Mitigating plasma turbulence in fusion reactors. *Nucl Fusion*. 2016;56(4): 046002. <https://doi.org/10.1088/0029-5515/56/4/046002>.
29. Volpe F. Recent advances in fusion reactor technology. *Fusion Sci Technol*. 2022;78(1):1–12. <https://doi.org/10.13182/FST21-1>.
30. Mitchell AC, et al. Materials for fusion reactors: challenges and innovations. *Mater Sci Eng*. 2021;815:141–50. <https://doi.org/10.1016/j.mse.2021.141150>.
31. Molodyk A, Samoilnikov S, Markelov A, et al. Development and large volume production of extremely high current density $\text{YBa}_2\text{Cu}_3\text{O}_7$ superconducting wires for fusion. *Sci Rep*. 2021;11:2084. <https://doi.org/10.1038/s41598-021-81559-z>.
32. International Atomic Energy Agency. Fusion energy: the key to a sustainable future. IAEA Reports. 2021.
33. Bennemann KH, Ketterson J. Helium: the inert gas and its significance in fusion reactors. *J Low-Temp Phys*. 2008;153(1–2):1–15. <https://doi.org/10.1007/s10909-008-9794-5>.
34. International Energy Agency. Fusion energy: a key player in the energy transition. IEA Insights. 2022. <https://www.iea.org/reports/fusion-energy-a-key-player-in-the-energy-transition>.
35. Tanna R. The tritium fuel cycle in fusion reactors. *Fusion Eng Des*. 2006;81(12):2391–5. <https://doi.org/10.1016/j.fusengdes.2006.07.017>.
36. Lawson JD. Some criteria for a power producing thermonuclear reactor. *Proc Phys Soc*. 1955;70(6):6–26. <https://doi.org/10.1088/0370-1328/70/6/301>.
37. South Korea KSTAR. Research on plasma confinement. *Nucl Fusion*. 2010;50(8): 084009. <https://doi.org/10.1088/0029-5515/50/8/084009>.

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.